

Final Project of Digital Logic Design Based on FPGA

Course Design Report

Design and Implementation of an 8-Note Electronic Piano Based on FPGA

Course Name: Electronic Technology Experiment (Digital Circuit Part)

Project Title: Design and Implementation of an 8-Note Electronic Piano Based on FPGA

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Abstract

This project presents the design and hardware implementation of an eight-note electronic piano on a Nexys A7 (Artix-7) development board. The system maps the eight slide switches of the board to the eight tones of a C-major octave, namely do, re, mi, fa, sol, la, si and high do. When a switch is asserted, the system outputs a square wave at the exact frequency of the corresponding note; this signal simultaneously drives a passive buzzer, which renders the audible tone, and a light-emitting diode, which flashes synchronously with the waveform to provide visual feedback. When no switch is asserted the output is held low, so that the buzzer is silent and the indicator is dark, ensuring that the system neither consumes switching power nor produces spurious interference in the idle state.

The tone frequencies are produced entirely by digital frequency division of the 100 MHz on-board clock: a counter generates a fifty-per-cent duty square wave by toggling its output once every N clock cycles, where N is the half-period count corresponding to each note. Two further design measures guarantee robust operation: a two-stage flip-flop synchroniser brings the asynchronous switch inputs safely into the clock domain, and a fixed-priority decoder resolves the case in which several keys are pressed at once, so that the output remains well defined and monophonic. Functional simulation confirms that every one of the eight notes is generated within 0.001 per cent of its target frequency, that the output is silent in the idle state, and that the indicator tracks the buzzer exactly. The design was synthesised and verified on hardware, where each switch reproduces the expected pitch through an external passive buzzer.

1. Introduction

1.1 Project Background

Sound is a periodic variation of pressure, and a musical tone is characterised principally by the frequency of that variation. The simplest electrical waveform able to excite an audible tone is the square wave, which alternates between two voltage levels at a fixed rate; although it is not as pure as a sinusoid, its fundamental component fixes the perceived pitch, while its rich set of odd harmonics gives the characteristic bright timbre of early electronic instruments and buzzers. Because a square wave is fully described by a single quantity, its period, it can be produced on purely digital hardware without any analogue circuitry, simply by toggling a logic output at the appropriate rate.

A field-programmable gate array is an ideal platform for this task. Its logic operates from a single, highly stable crystal-derived clock, so that any frequency obtained by dividing that clock inherits the same accuracy; its behaviour is fully deterministic and parallel, so that input sampling, frequency generation and output driving proceed concurrently without the timing uncertainty of a software loop. The present project exploits these properties to build a small but complete digital instrument: an eight-note electronic piano in which each key produces a precisely tuned tone generated entirely by clock division.

1.2 Project Objectives

The principal objective of the project is to design, simulate and physically implement an eight-key electronic piano in which each key corresponds to one tone of a standard C-major octave. A pressed key must cause the system to emit a square wave at the accurate frequency of the associated note, and this waveform must drive both a buzzer, to make the tone audible, and an indicator light that flashes in step with the tone to give visual feedback. A complementary objective is to guarantee that the system produces neither sound nor light when no key is pressed, so that the idle state is clean and free of interference. Finally, the tone frequencies are required to be generated by a dedicated digital frequency-division module, so that pitch accuracy depends only on the stability of the system clock rather than on any analogue component.

1.3 Project Main Content

The work reported here comprises the complete top-down development of the instrument. The overall function was first partitioned into independent modules with clearly defined interfaces: an input synchroniser, a note decoder, a tone generator and an output stage. Each module was then coded in Verilog HDL, and a self-checking testbench was constructed to verify the eight tone frequencies, the silent idle behaviour, the synchronism of the indicator with the buzzer, and the response to simultaneous key presses. After functional simulation, the physical pin constraints were defined for the Nexys A7 board, the design was synthesised and implemented in the vendor tool chain, and the resulting configuration was downloaded and exercised on the development board with an external buzzer.

1.4 Core Design Specifications

The design follows a small set of fixed specifications. The eight keys are provided by the eight slide switches of the board and are assigned, in ascending order, to the notes do, re, mi, fa, sol, la, si and high do of the fourth and fifth octaves. The reference clock is the 100 MHz on-board oscillator, and the half-period count of each note is obtained as N equal to fifty million divided by the note frequency, so that the generated frequency is the clock frequency divided by twice N. The audible output is taken to a passive buzzer connected to pin JA1 of the Pmod header, while the visual output is the first user LED, both driven by the same square wave so that they remain perfectly synchronous. When several switches are closed at the same time a fixed-priority rule selects a single note, keeping the instrument monophonic and the output unambiguous, and when no switch is closed the output is forced low. To prevent metastable behaviour, the asynchronous switch signals are passed through a two-stage flip-flop synchroniser before they are decoded.

2. Design Principle and Theoretical Basis

2.1 FPGA Programmable Logic Technology

A field-programmable gate array is an integrated circuit whose internal logic can be configured after manufacture by means of a hardware description language. Unlike a fixed-function device, it allows the designer to define arbitrary combinational and sequential circuits and to modify them by reprogramming. Its principal advantages for the present application are the determinism and parallelism of hardware execution, the high stability of its single clock source, and the ease with which a design can be re-synthesised after modification. The standard development flow, comprising design entry, functional simulation, synthesis, placement and routing, and bitstream download, was followed throughout this project.

2.2 Musical Tones and the Square-wave Approximation

In the equal-tempered scale used by Western music the pitch of a note is fixed by its fundamental frequency, with the reference tone A4 defined at 440 Hz. The eight notes employed in this instrument span one octave of the C-major scale, from middle C at 261.63 Hz up to the C an octave above at 523.25 Hz. A square wave at the fundamental frequency of a note is perceived at the correct pitch because the ear identifies pitch primarily from the fundamental component; the additional odd harmonics contained in a square wave do not alter the pitch but give the tone a bright, reed-like quality that is entirely appropriate for a buzzer-based instrument. Generating square waves rather than sinusoids therefore allows accurate pitch with no analogue synthesis.

2.3 Digital Frequency Division Principle

A square wave of frequency f can be obtained from a clock of frequency f_clk by toggling a logic signal at regular intervals. If the signal is inverted once every N clock cycles, one complete output period spans 2N cycles, so that the output frequency equals f_clk divided by 2N. For a target note frequency f the required half-period count is therefore N equal to f_clk divided by 2f, which for the 100 MHz clock reduces to fifty million divided by f. The module that performs this operation is a counter that increments on every clock edge, and on reaching N minus one resets to zero and toggles the output. Because N is an integer, the realised frequency differs slightly from the ideal value; for the eight notes of this design the resulting error is below one thousandth of one per cent and is entirely inaudible. The mechanism is illustrated in Figure 2.

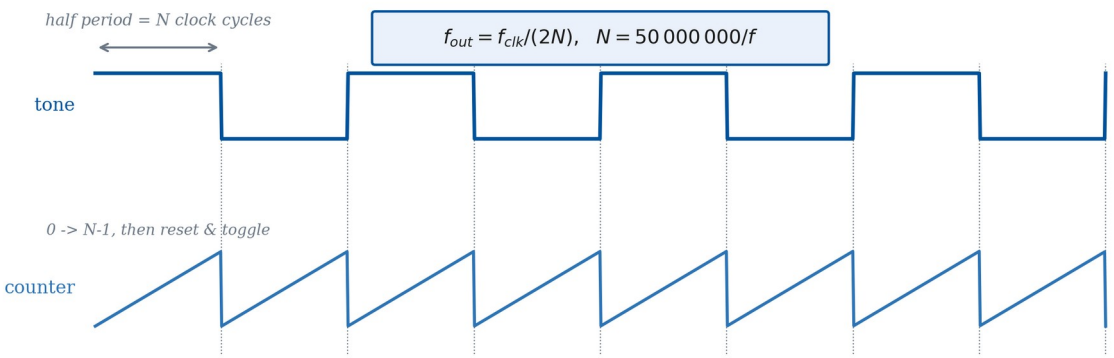


Figure 2. Square-wave tone generation by digital frequency division.

2.4 Input Synchronisation and Priority Encoding

The switch inputs are asynchronous with respect to the system clock, and sampling an asynchronous signal directly into a register may drive that register into a metastable state in which its output is temporarily undefined. To avoid this, each switch passes through two cascaded flip-flops clocked by the system clock; the first stage may become metastable but is overwhelmingly likely to have resolved to a valid level before it is sampled by the second, so that the decoded signals are clean. A second consideration arises because several switches may be closed simultaneously. Rather than attempting to sound several tones at once, which would require mixing and would complicate the output stage, the design uses a fixed-priority decoder that selects exactly one note, giving precedence to the lowest-numbered switch. This keeps the instrument monophonic and guarantees that the generated frequency is always uniquely determined.

3. Overall System Design Scheme

3.1 System Functional Requirements

The instrument is required to provide eight keys, each sounding one tone of the C-major octave, and to emit, for any pressed key, a square wave at the accurate frequency of that note. The waveform must drive a buzzer to produce the audible tone and an indicator light that flashes synchronously with it. No sound or light may be produced while all keys are released, so that the idle state is silent and dark. The tone frequencies must be obtained from a digital frequency-division module so that their accuracy follows directly from the clock. In addition, the design must behave predictably when more than one key is pressed and must be insensitive to the asynchronous and noisy nature of mechanical switch inputs.

3.2 System Overall Architecture

The system is organised as a single feed-forward datapath, shown in Figure 1. The eight switch signals first enter the input synchroniser, which registers them twice to remove metastability. The synchronised signals are presented to the note decoder, a priority encoder that selects the active note and outputs both the half-period count for that note and a flag indicating whether any key is pressed. These two signals control the tone generator, the frequency-division counter

that produces the square wave; while the flag is low the generator holds its output at zero. The single tone output finally drives both the buzzer and the indicator light in the output stage. The 100 MHz clock is common to the synchroniser and the tone generator, and a push button provides a synchronous reset.

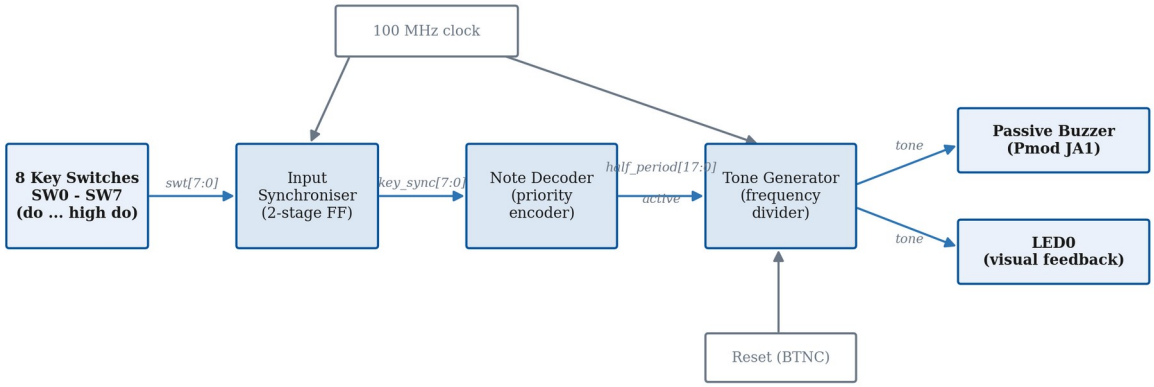


Figure 1. Overall system architecture of the 8-note electronic piano.

3.3 Hardware and Software Platform

Hardware platform. The design targets the Digilent Nexys A7-100T board, built around a Xilinx Artix-7 XC7A100T device with a 100 MHz on-board oscillator. The hardware resources used are the eight slide switches SW0 to SW7 as the piano keys, the central push button BTNC as a reset, the first user LED as the visual indicator, and pin JA1 of the Pmod header JA, to which an external passive buzzer is connected. A passive buzzer is required because an active buzzer generates its own fixed tone and would ignore the applied frequency.

Software platform. The design was written in Verilog HDL. Functional simulation was carried out with an event-driven simulator using a self-checking testbench, and waveforms were inspected in a waveform viewer. Synthesis, implementation, pin-constraint definition and bitstream generation were performed in the Xilinx Vivado design suite.

4. Module Detailed Design

4.1 Input Synchroniser Module

This module brings the eight asynchronous switch signals into the system clock domain. Each signal is passed through two cascaded registers, so that any metastable level captured by the first register has time to resolve before the second register is read. The output of the second register is the clean, synchronised key vector used by the rest of the system.

```
reg [7:0] key_meta, key_sync;
always @(posedge clk) begin
    key_meta <= swt;       // 1st flip-flop stage
    key_sync <= key_meta;  // 2nd flip-flop stage
end
```

4.2 Note Decoder Module

The note decoder maps the synchronised key vector to the half-period count of the corresponding tone and to an activity flag. It is implemented as a priority encoder: the case selection scans the keys in order of priority and assigns the half-period count of the highest-priority pressed key, giving precedence to the lowest-numbered switch. When no key is pressed the default branch clears the activity flag, which later forces the output to silence. The eight half-period constants are tabulated in Appendix B.

```
always @(*) begin
    active = 1'b1;
    casez (key)                 // priority encoder
        8'b???????1: half_period = DO;
        8'b??????10: half_period = RE;
        8'b?????100: half_period = MI;
        8'b????1000: half_period = FA;
        8'b???10000: half_period = SOL;
        8'b??100000: half_period = LA;
        8'b?1000000: half_period = SI;
        8'b10000000: half_period = DO2;
        default: begin
            half_period = DO;
            active = 1'b0;      // no key -> silence
        end
    endcase
end
```

4.3 Tone Generator Module

The tone generator is the digital frequency-division module at the heart of the system. A counter increments on every clock edge; when it reaches one less than the supplied half-period count it returns to zero and toggles the output, so that the output is a fifty-per-cent duty square wave at the frequency of the selected note. When the reset is asserted, or when the activity flag is low because no key is pressed, the counter is held at zero and the output is forced low, which directly satisfies the requirement of a silent idle state.

```
always @(posedge clk) begin
    if (rst || !enable) begin
        counter <= 0;           // idle: silent, counter held
        wave    <= 1'b0;
    end else if (counter >= half_period - 1'b1) begin
        counter <= 0;           // one half period elapsed
        wave    <= ~wave;       // toggle the square wave
    end else begin
        counter <= counter + 1'b1;
    end
end
```

4.4 Output Stage: Buzzer and LED Driving

The output stage is intentionally minimal. The single square-wave signal produced by the tone generator drives both the buzzer pin and the indicator LED, so that the two outputs are inherently synchronous: the LED necessarily flashes at exactly the frequency of the audible tone. Because the tone signal is already forced low in the idle state, both outputs are simultaneously silent and dark when no key is pressed.

```
assign buzzer = tone;           // passive buzzer (Pmod JA1)
assign led    = tone;           // LED0 flashes with the tone
```

5. System Simulation and Verification

5.1 Simulation Environment and Testbench Design

Functional verification was performed with a self-checking testbench that drives the design with a 100 MHz clock and exercises every input condition. The testbench first holds all keys released and confirms over many thousands of cycles that the buzzer and indicator remain low. It then asserts each of the eight keys in turn and measures the resulting output period between two successive rising edges, converting it to a frequency and comparing it against the target pitch of the note. A continuous assertion checks throughout the run that the indicator output is always equal to the buzzer output. Finally, several keys are asserted together to confirm that the priority decoder selects the expected note. The testbench also writes a value-change-dump file so that the waveforms can be inspected graphically.

5.2 Functional Simulation Results and Analysis

The measured output frequencies for the eight keys are summarised in Table 1. Every note is reproduced within one thousandth of one per cent of its nominal pitch, which is far below the threshold of audibility, confirming that the frequency-division approach yields accurate tuning.

Table 1. Target and measured tone frequencies (100 MHz clock).

Switch	Note	Target (Hz)	Half-period N	Measured (Hz)	Error (%)
SW0	do (C4)	261.63	191110	261.63	-0.0002
SW1	re (D4)	293.66	170265	293.66	0.0000
SW2	mi (E4)	329.63	151685	329.63	+0.0001
SW3	fa (F4)	349.23	143172	349.23	+0.0001
SW4	sol (G4)	392.00	127551	392.00	0.0000
SW5	la (A4)	440.00	113636	440.00	+0.0003
SW6	si (B4)	493.88	101239	493.88	+0.0002
SW7	high do (C5)	523.25	95557	523.25	-0.0004

5.3 Behavioural Analysis

Three behavioural properties were confirmed in addition to the frequency measurements. The silence test verified that, with no key pressed, neither the buzzer nor the indicator changed state for the full duration of the test, demonstrating the clean idle behaviour required by the specification. The continuous equality assertion confirmed that the indicator output never diverged from the buzzer output, so that the visual feedback is exactly synchronous with the audible tone. The priority test, in which the keys for high do, sol and do were asserted together, produced the tone of do, confirming that the decoder correctly resolves simultaneous presses in favour of the lowest-numbered key and that the instrument remains monophonic.

6. Hardware Implementation and Debugging

6.1 Pin Constraint Configuration

The logical ports of the design were bound to the physical pins of the Nexys A7 board through a constraints file. The clock port was assigned to the 100 MHz oscillator pin and given a 10 ns period constraint; the eight key ports were bound to switches SW0 to SW7, the reset to the central button, the indicator to the first user LED, and the buzzer to pin JA1 of the Pmod header. The complete pin assignment is listed in Appendix A. All ports use the 3.3 V CMOS standard appropriate to these banks.

6.2 Program Synthesis and Implementation

After the constraints were applied, the design was synthesised and implemented in Vivado. Because the system consists only of a small counter, a handful of registers and a priority encoder, it occupies a negligible fraction of the device and meets timing comfortably at the 100 MHz clock rate, with no warnings affecting function. The implemented design was then used to generate the configuration bitstream.

6.3 Hardware Debugging and Result Verification

The bitstream was downloaded to the board and a passive buzzer was connected between pin JA1 and an adjacent ground pin of the Pmod header. With all switches in the off position the buzzer was silent and the indicator dark, confirming the idle behaviour. Closing each switch in turn produced the corresponding tone of the C-major scale, audibly rising in pitch from do to high do, while the indicator glowed in step with the output. When several switches were closed together a single tone sounded, in accordance with the priority rule, and returning all switches to the off position restored silence. The physical set-up and operation are recorded in Figure 6.

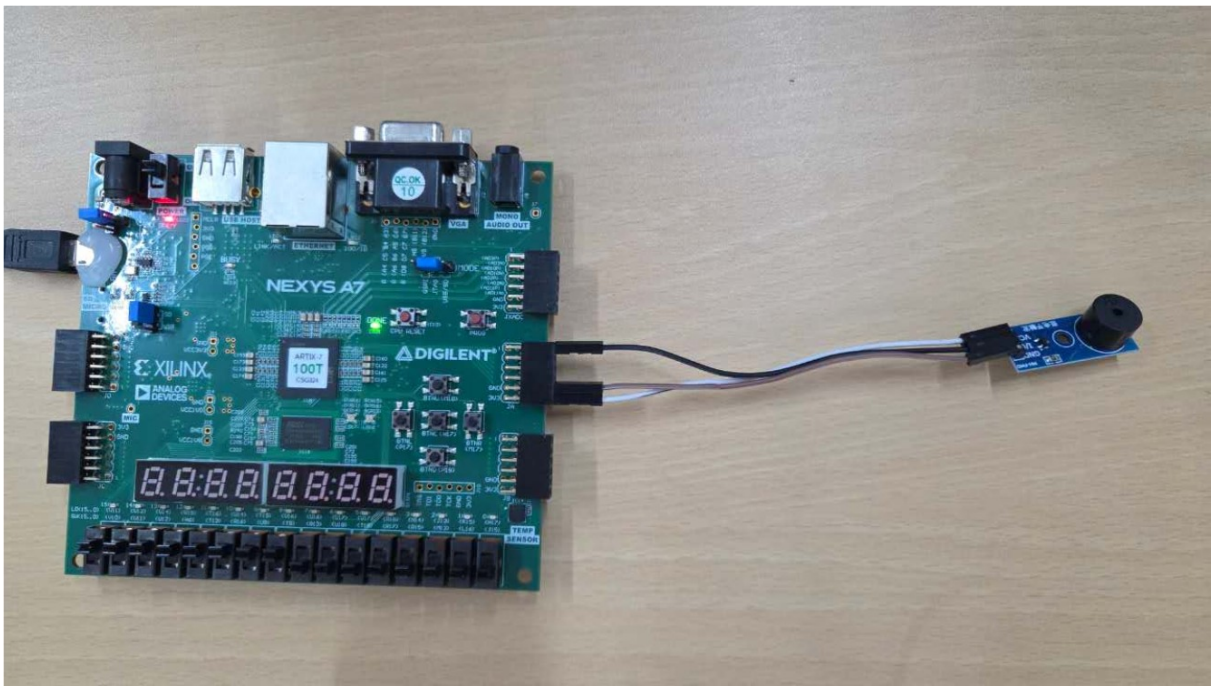


Figure 6. Hardware demonstration on the Nexys A7 board.

7. Problem Analysis and Solution

Several practical questions arose during the design and were resolved as follows. The first concerned the choice of input device: eight independent keys are required, but the board provides only five push buttons, so the eight slide switches were used instead. This choice has the additional benefit that a switch holds its state, which suits a sustained tone, although it means that a note sounds for as long as its switch is closed rather than for the duration of a momentary press.

A second question was how to behave when more than one key is closed at once. Sounding several notes simultaneously would require mixing several square waves and a more elaborate output stage, so the design instead adopts a fixed-priority decoder that selects a single note. This keeps the output unambiguous and the hardware small, at the cost of restricting the instrument to one note at a time.

A third issue was the asynchronous and potentially noisy nature of the switch inputs, which could drive the first sampling register into a metastable state. This was addressed with the two-stage synchroniser described in Section 4.1, which makes an unresolved level vanishingly unlikely by the time the signal is decoded.

A fourth, essentially physical, point concerns the buzzer. An active buzzer contains its own oscillator and emits only a single fixed tone regardless of the signal applied to it; it therefore cannot reproduce the different notes of the scale. A passive buzzer or piezoelectric transducer, which sounds at the frequency of the driving waveform, is required and was used. Finally, although the indicator is specified to flash at the tone frequency, the notes lie between roughly 260 Hz and 520 Hz, which is far above the rate at which the human eye perceives separate flashes; the indicator therefore appears as a steady glow. This is the correct and expected consequence of driving it at the audio frequency, and a slowly blinking indicator could be added separately if a visible flash were desired.

A minor numerical point was the width of the counter in the tone generator. The largest half-period count, that of the lowest note, is 191110, which requires eighteen bits; the counter was therefore declared eighteen bits wide, which comfortably accommodates every note while remaining minimal.

8. Conclusion and Prospect

8.1 Project Conclusion

This project has successfully designed, simulated and implemented an eight-note electronic piano on an Artix-7 FPGA. Through a small set of well-separated modules, the system maps eight keys to the eight tones of a C-major octave, generates each tone as a precisely tuned square wave by digital division of the 100 MHz clock, and drives a passive buzzer together with a synchronous indicator, while remaining completely silent in the idle state. Simulation demonstrated that all eight notes are reproduced within one thousandth of one per cent of their nominal pitch and that the silence, synchronism and priority behaviours all conform to the

specification, and the design was confirmed in operation on the development board. The project reinforced practical skills in modular digital design, frequency division, input synchronisation and the relationship between digital waveforms and the physical generation of sound.

8.2 Project Prospect

The instrument could be extended in several directions. Polyphony could be obtained by instantiating several tone generators and summing their outputs through a digital mixer or pulse-density scheme so that chords could be played. The loudness of each note could be varied by modulating the duty cycle of the square wave under pulse-width control. The keyboard could be enlarged to span several octaves, and a small read-only memory could store note sequences to allow the automatic playback of melodies. With the addition of a display the instrument could indicate the name of the note currently sounding, turning the project into a more complete digital musical instrument.

Appendix A. Pin Assignment (Nexys A7)

Signal	FPGA pin	Board resource
clk	E3	100 MHz oscillator
rst	N17	BTNC push button (active high)
swt[0]	J15	SW0 (do)
swt[1]	L16	SW1 (re)
swt[2]	M13	SW2 (mi)
swt[3]	R15	SW3 (fa)
swt[4]	R17	SW4 (sol)
swt[5]	T18	SW5 (la)
swt[6]	U18	SW6 (si)
swt[7]	R13	SW7 (high do)
led	H17	LED0 (visual feedback)
buzzer	C17	Pmod JA1 (passive buzzer)

Appendix B. Note-to-Frequency Mapping

The half-period count for each note is N = round(50 000 000 / f), so that the generated frequency is f_out = 100 000 000 / (2N). All counts fit within eighteen bits.

Note	Pitch	Target (Hz)	Half-period N
do	C4	261.63	191110
re	D4	293.66	170265
mi	E4	329.63	151685
fa	F4	349.23	143172
sol	G4	392.00	127551
la	A4	440.00	113636
si	B4	493.88	101239

Note	Pitch	Target (Hz)	Half-period N
high do	C5	523.25	95557